

DABAA, Egypt

WMO: 623090

Lat: 31.0327N

Lon: 28.4448E

Elev: 18

StdP: 101.11

Time Zone: 2.00 (E02)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	8.6	-2.7	3.0	16.9	-0.2	3.7	15.6	18.1	11.0	13.6	12.8	3.3	240	0.351

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	33.5	21.0	31.8	22.9	31.0	23.4	25.5	29.8	25.0	29.4	24.5	29.0	3.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.2	27.9	23.6	18.5	27.8	23.1	17.9	27.6	78.3	29.8	76.3	29.3	74.5	29.1	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.6	9.1	7.5	DB	5.9	40.0	0.9	2.9	5.3	42.1	4.8	43.8	4.3	45.4	3.6	47.5
			WB	3.3	26.6	1.4	1.0	2.2	27.3	1.4	27.8	0.6	28.4	-0.5	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	14.0	14.9	16.7	18.8	21.7	24.7	26.8	27.5	26.3	23.5	19.9	16.0
	DBStd	5.07	1.92	2.46	2.64	2.48	2.49	1.80	1.31	1.07	1.41	1.81	2.10	1.94
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	407	134	102	67	19	1	0	0	0	0	0	8	77
	CDD10.0	3991	125	137	206	265	363	440	522	542	489	419	297	186
	CDD18.3	1356	1	4	14	34	106	190	263	283	239	162	56	3
	CDH23.3	11716	9	42	127	214	586	1468	2711	3175	2274	949	151	10
	CDH26.7	3566	1	13	49	77	180	369	888	1138	671	166	13	2
Wind	WSAvg	3.1	3.5	3.7	3.5	3.4	3.2	3.1	3.3	2.8	2.7	2.5	2.4	3.0
Precipitation	PrecAvg	173	44	31	16	6	2	0	0	0	1	8	22	42
	PrecMax	273	83	68	42	20	12	3	0	1	6	30	86	91
	PrecMin	90	17	8	1	0	0	0	0	0	0	0	2	9
	PrecStd	41	19	15	10	5	3	0	0	0	2	7	18	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	28.5	32.2	33.1	36.0	35.1	35.2	34.2	34.9	32.7	28.2	24.4
		MCWB	13.7	14.9	16.0	16.7	18.8	20.4	22.5	23.5	21.6	20.0	18.4	15.5
	2%	DB	20.9	23.6	26.7	28.8	30.9	31.1	32.0	32.0	31.8	29.7	26.2	21.9
		MCWB	13.0	13.1	14.7	16.1	18.4	21.7	23.9	24.5	23.1	20.9	19.3	15.6
	5%	DB	19.2	20.7	23.1	25.5	28.0	29.7	31.0	31.3	30.6	28.2	25.0	20.9
		MCWB	13.2	12.8	14.5	16.3	19.0	22.2	23.8	24.3	23.3	21.3	18.5	15.1
	10%	DB	18.2	19.0	20.8	23.1	26.0	28.6	30.2	30.7	29.7	27.2	23.9	20.0
		MCWB	12.9	13.1	14.6	16.5	19.2	22.1	23.6	24.1	23.0	20.9	18.1	14.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.9	18.0	19.2	21.7	24.5	25.9	26.4	25.5	23.9	22.2	18.7
		MCDB	19.8	21.0	24.2	24.7	26.6	28.8	29.9	30.6	30.0	27.8	25.4	22.2
	2%	WB	14.9	15.6	16.8	18.4	21.0	23.9	25.2	25.7	24.7	22.9	20.8	16.8
		MCDB	18.7	19.6	21.4	23.6	25.7	28.0	29.4	29.9	29.2	27.3	24.5	20.8
	5%	WB	14.1	14.7	16.0	17.7	20.5	23.4	24.7	25.2	24.1	22.1	19.8	15.8
		MCDB	18.1	18.7	20.6	22.5	25.0	27.5	29.0	29.6	28.8	26.6	23.8	19.7
	10%	WB	13.4	13.9	15.3	17.1	19.9	22.8	24.2	24.7	23.5	21.5	18.7	15.1
		MCDB	17.4	18.1	19.6	21.7	24.4	26.9	28.6	29.2	28.4	26.1	22.8	19.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.3	8.4	8.4	8.5	7.7	7.1	7.1	7.3	7.4	7.6	7.9
		MCDBR	10.1	11.8	13.1	13.3	13.0	9.5	8.4	8.0	8.9	9.2	9.2	9.2
		MCWBR	5.6	5.3	5.3	4.7	4.5	3.4	3.0	3.1	3.6	4.1	4.6	5.2
	5% WB	MCDBR	8.9	8.8	9.4	9.6	9.1	7.6	7.8	7.5	7.8	8.1	8.2	8.3
		MCWBR	5.6	4.9	4.7	4.2	3.8	3.0	3.0	3.0	3.6	4.1	4.5	5.4
Clear-Sky Solar Irradiance	taub		0.391	0.439	0.467	0.513	0.503	0.439	0.423	0.427	0.442	0.427	0.396	0.389
	taud		2.165	2.007	1.932	1.841	1.896	2.118	2.173	2.169	2.127	2.164	2.218	2.192
	Ebn at Noon		814	806	813	792	801	851	863	856	828	810	804	793
	Edh at Noon		123	159	183	208	198	158	149	148	149	134	116	115
All-Sky Solar Radiation	RadAvg		3.18	4.04	5.37	6.51	7.28	7.83	7.69	7.02	6.04	4.70	3.49	2.89
	RadStd		0.14	0.23	0.20	0.19	0.22	0.13	0.09	0.10	0.11	0.16	0.14	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.42	N/A	N/A	N/A	N/A	+0.25	N/A	-62	+171	+104

Nomenclature: See separate page